

화학수학

일곱 번째 수업

급수

Line Integral (선적분)

Consider a differential with two independent variables:

$$dF = M(x, y)dx + N(x, y)dy$$

For the curve C joining (x_0, y_0) and (x_1, y_1) ,
the **line integral** (path integral) is defined as

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

Evaluation of Line Integral

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C dF = \int_C [M\{x, y(x)\}dx + N\{x(y), y\}dy]$$

$$\int_C dF = \int_{x_0}^{x_1} M\{x, y(x)\} dx + \int_{y_0}^{y_1} N\{x(y), y\} dy$$

Line Integrals of Exact Differentials

If dF is an exact differential, the line integral $\int_C dF$

- depends **only on the initial and final points**.
- equals to **$F(\text{final point}) - F(\text{initial point})$** .

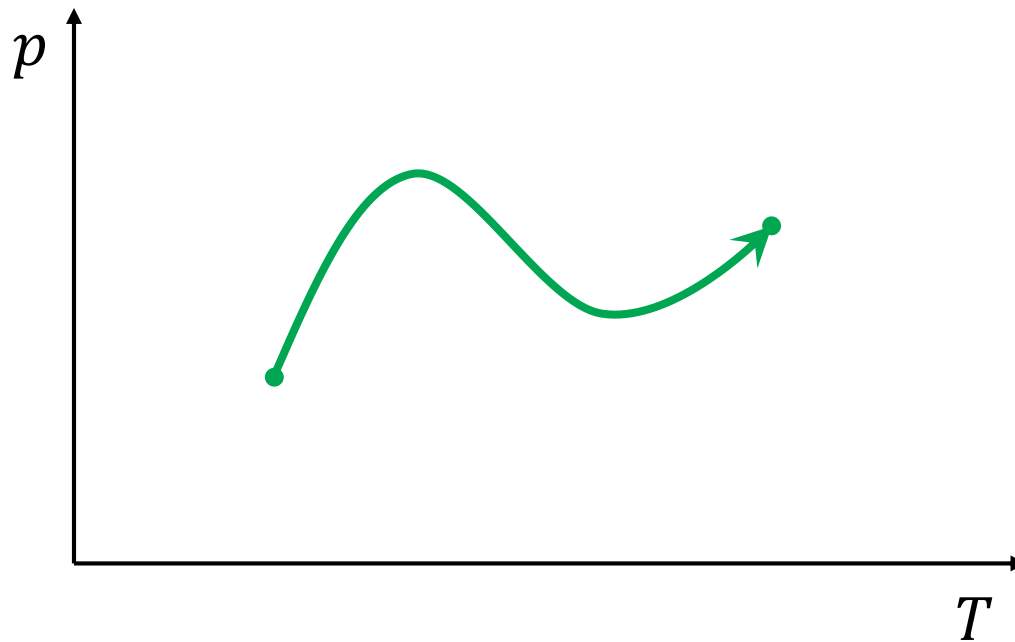
For an exact differential dF of the function $F(x, y)$,
with the initial point (x_0, y_0) and the final point (x_1, y_1) ,

$$\int_C dF = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y dx + \left(\frac{\partial F}{\partial y} \right)_x dy \right] = F(x_1, y_1) - F(x_0, y_0)$$



Line Integrals in Thermodynamics

- Equilibrium state: a point in a space
- Reversible process: a line integral



Double Integral (이중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(x, y) \, dy \, dx$$


- Carry out **the inner integration** first
 - Treat the other independent variable (x) as a constant
- Then carry out **the outer integration**

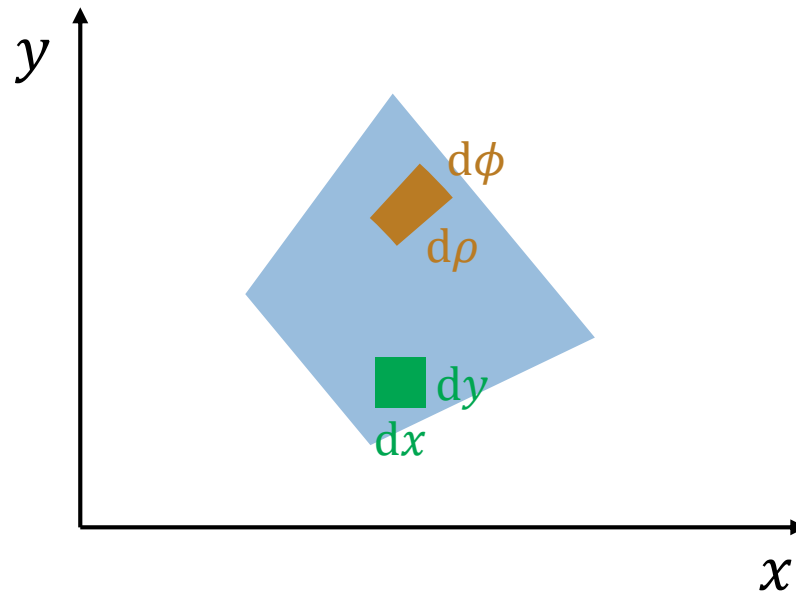
Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

- First, integrate z from c_1 to c_2 ($x, y = \text{constant}$)
- Then integrate y from b_1 to b_2 ($x = \text{constant}$)
- Lastly, integrate x from a_1 to a_2

In Different Coordinates

$$\iint f \, dA = \iint f(x, y) \, dx \, dy = \iint f(\rho, \phi) \rho \, d\rho \, d\phi$$



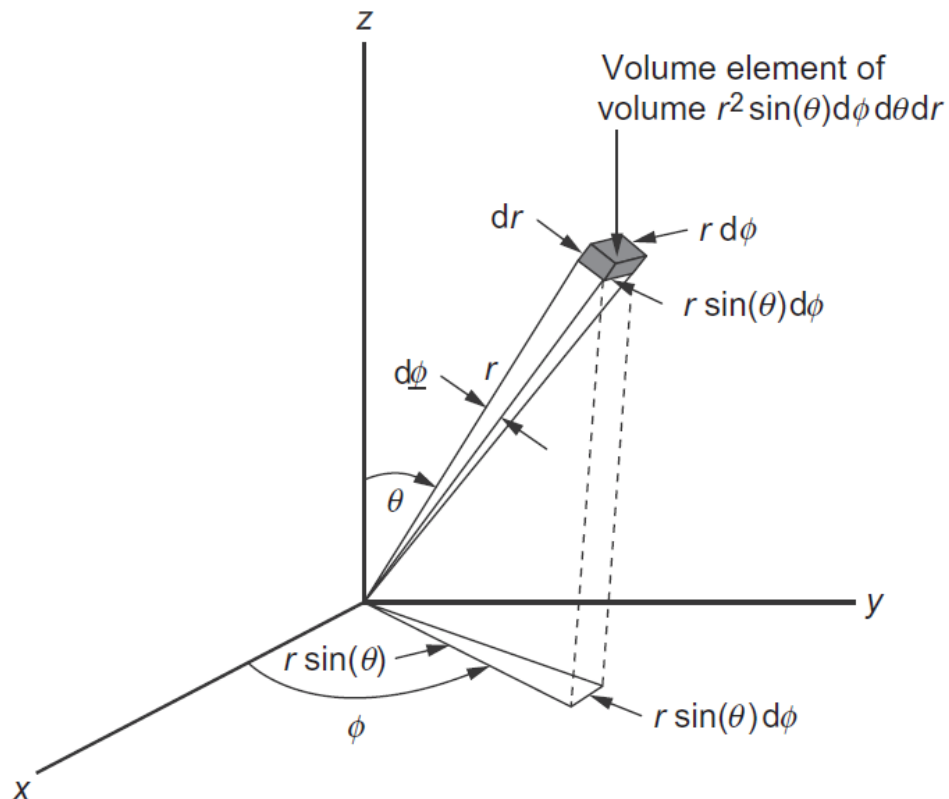
Jacobian

$$\int \int f(x, y) \, dx \, dy = \int \int f(\rho, \phi) \frac{\partial(x, y)}{\partial(\rho, \phi)} \, d\rho \, d\phi$$

where

$$\frac{\partial(x, y)}{\partial(\rho, \phi)} = \begin{vmatrix} \partial x / \partial \rho & \partial x / \partial \phi \\ \partial y / \partial \rho & \partial y / \partial \phi \end{vmatrix} = \left(\frac{\partial x}{\partial \rho} \right) \left(\frac{\partial y}{\partial \phi} \right) - \left(\frac{\partial x}{\partial \phi} \right) \left(\frac{\partial y}{\partial \rho} \right)$$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin \theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Today's Class

- Mathematical series
 - Series: geometric series
 - Functional series
- Power series
 - Maclaurin series
 - Taylor series

10. Mathematical Series

(10장 급수)

Sequence and Series

- **Sequence** (수열): $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_n \rightarrow \cdots$
 - Rule for generating one member from previous one
- **Series** (급수): $a_0 + a_1 + a_2 + \cdots$
 - Finite series (유한급수): $s = a_0 + a_1 + \cdots + a_n$
 - Infinite series (무한급수): $s = a_0 + a_1 + \cdots + a_n + \cdots$

More about Infinite Series

$$s = a_0 + a_1 + \cdots + a_n + \cdots = \sum_{k=0}^{\infty} a_k$$

- n th partial sum (부분합): $S_n = a_0 + a_1 + \cdots + a_{n-1}$
 - $s = \lim_{n \rightarrow \infty} S_n$
- If s exists and is finite, the series **converges**; otherwise, the series **diverges**.

Geometric Series (기하급수)

$$s = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = a \sum_{n=0}^{\infty} r^n$$

- Convergence condition: $|r| < 1$
- Value: since $s = a + r(a + ar + ar^2 + \cdots) = a + rs$,

$$s = \frac{a}{1 - r}$$

Example 10.3

The molecular partition function z is defined as

$$z = \sum_{i=0}^{\infty} \exp\left(-\frac{E_i}{k_B T}\right).$$

If we consider only the vibration of a diatomic molecule, to a good approximation,

$$E_i = E_v = h\nu \left(v + \frac{1}{2}\right).$$

Find an expression of the vibrational partition function.

Example 10.3

$$z = \sum_{v=0}^{\infty} \exp\left(-\frac{E_v}{k_B T}\right), \quad E_v = h\nu\left(v + \frac{1}{2}\right)$$

Substitution leads to

$$z = \sum_{v=0}^{\infty} \exp\left[-\frac{h\nu}{k_B T}\left(v + \frac{1}{2}\right)\right] = e^{-h\nu/2k_B T} \sum_{v=0}^{\infty} e^{-h\nu v/k_B T}$$

Introduce $r = e^{-h\nu/k_B T}$ to yield

$$z = r^{1/2} \sum_{v=0}^{\infty} r^v = r^{1/2} \frac{1}{1-r} = \frac{e^{-h\nu/2k_B T}}{1 - e^{-h\nu/k_B T}}$$

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

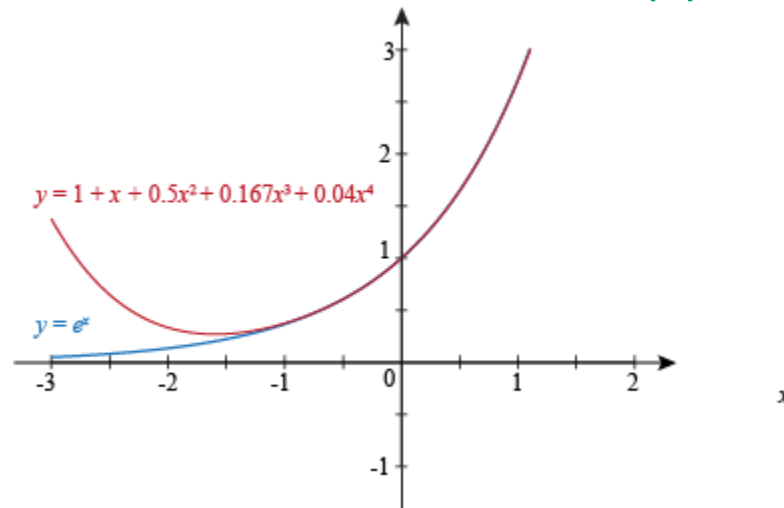
- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Approximation to a Function

- Functional series can be used to represent a function

$$s(x) = a_0g_0(x) + a_1g_1(x) + a_2g_2(x) + \cdots \rightarrow f(x)$$

- Its partial sum can be used as an approximation



(Image: <https://www.shmoop.com/series/taylor-maclaurin-series-examples.html>)

Power Series

- Maclaurin Series

$$s(x) = a_0 + a_1x + a_2x^2 + \dots$$

- Taylor Series

$$s(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

Maclaurin Series to Represent $f(x)$

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

At $x = 0$, $f(0) = a_0$ and

$$\left(\frac{d^n f}{dx^n} \right)_{x=0} = n! a_n$$

Hence,

$$a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

Example 10.7

Find all of the coefficients for the Maclaurin series representing $\sin x$.

Consider $x = 0$. Since $\sin 0 = 0$, $a_0 = 0$.

For $f(x) = \sin x$,

$$\frac{df}{dx} = \cos x, \quad \frac{d^2f}{dx^2} = -\sin x, \quad \frac{d^3f}{dx^3} = -\cos x, \quad \frac{d^4f}{dx^4} = \sin x, \dots$$

Hence, using $a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$,

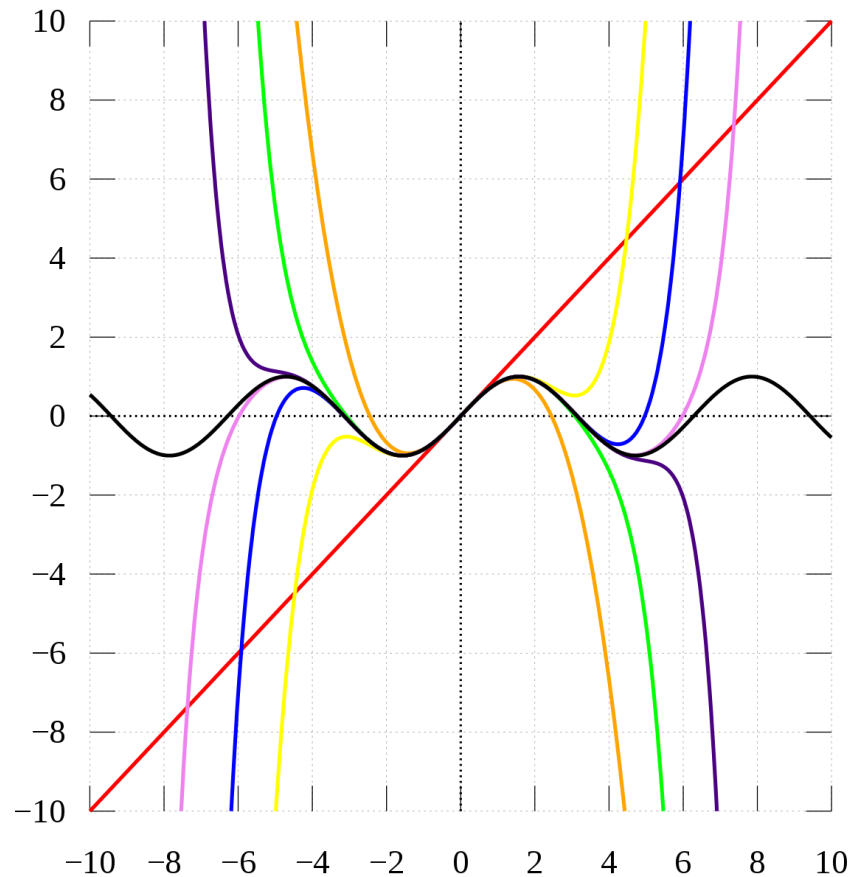
$$a_1 = 1, \quad a_2 = 0, \quad a_3 = -1/3!, \quad a_4 = 0, \dots$$

and

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

Not in Mortimer

Maclaurin Approximation to Sine



(Image: https://commons.wikimedia.org/wiki/File:Sintay_SVG.svg)

Taylor Series to Represent $f(x)$

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

where $h \neq 0$.

$$a_0 = f(h), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=h}$$

- Often called **Taylor expansion about $x = h$**
- A function that is not differentiable at $x = 0$ can be represented by a Taylor series

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

The series is given as

$$\ln x = a_0 + a_1(x - 1) + a_2(x - 1)^2 + a_3(x - 1)^3 + a_4(x - 1)^4 \dots$$

where

$$a_0 = f(1), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=1}$$

For $f(x) = \ln x$, $a_0 = \ln 1 = 0$.

Also,

$$\frac{df}{dx} = \frac{1}{x}, \quad \frac{d^2 f}{dx^2} = -\frac{1}{x^2}, \quad \frac{d^3 f}{dx^3} = \frac{2}{x^3}, \quad \frac{d^4 f}{dx^4} = -\frac{3!}{x^4}, \dots$$

Example 10.8

Find the Taylor series for $\ln x$, expanding about $x = 1$.

$$\frac{df}{dx} = \frac{1}{x}, \quad \frac{d^2f}{dx^2} = -\frac{1}{x^2}, \quad \frac{d^3f}{dx^3} = \frac{2}{x^3}, \quad \frac{d^4f}{dx^4} = -\frac{3!}{x^4}, \dots$$

Hence,

$$\begin{aligned} a_1 &= \frac{1}{1!} \left(\frac{df}{dx} \right)_{x=1} = 1, & a_2 &= \frac{1}{2!} \left(\frac{d^2f}{dx^2} \right)_{x=1} = -\frac{1}{2}, \\ a_3 &= \frac{1}{3!} \left(\frac{d^3f}{dx^3} \right)_{x=1} = \frac{1}{3}, & a_4 &= \frac{1}{4!} \left(\frac{d^4f}{dx^4} \right)_{x=1} = -\frac{1}{4}, \dots \end{aligned}$$

Therefore,

$$\ln x = (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3 - \frac{1}{4}(x - 1)^4 + \dots$$

Change of Variable

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

can be written as

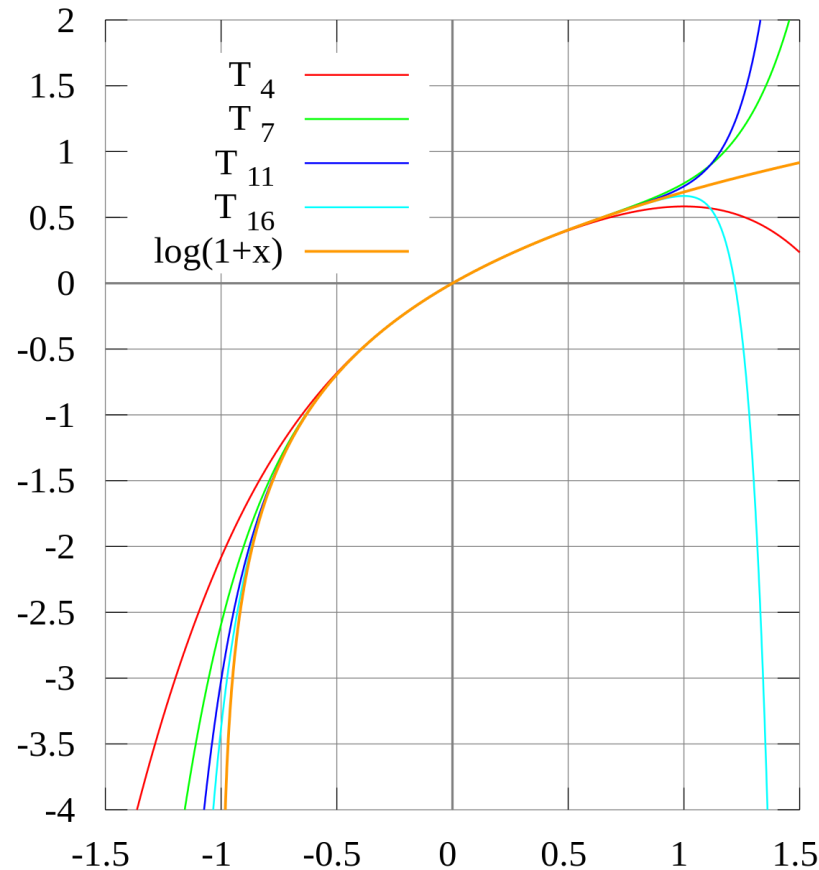
$$f(x + h) = a_0 + a_1x + a_2x^2 + \dots$$

$$a_0 = f(x = 0), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

e.g. Taylor expansion of $\ln(x + 1)$:

$$\ln(x + 1) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

Taylor Approximation to $\ln(x + 1)$



(Image: <https://commons.wikimedia.org/wiki/File:LogTay.svg>)

Mathematical Operations on Series

Differentiation/integration of a series?

For a (uniformly convergent) functional series

$$f(x) = \sum_{n=0}^{\infty} a_n g_n(x),$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] = \sum_{n=0}^{\infty} a_n \frac{dg_n(x)}{dx}$$

$$\int_b^c f(x) dx = \int_b^c \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] dx = \sum_{n=0}^{\infty} a_n \int_b^c g_n(x) dx$$

Example 10.12

Find the Maclaurin series for $\cos x$ from the Maclaurin series for $\sin x$.

The Maclaurin series for $\sin x$ is

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

Since $d(\sin x)/dx = \cos x$, differentiating both sides,

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ + a_{21}x^2y + a_{12}xy^2 + \dots$$

$$x = 0, y = 0 \text{ gives } f(0, 0) = a_{00}$$

What about other coefficients?

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ + a_{21}x^2y + a_{12}xy^2 + \dots$$

$$(\partial f / \partial x)_y = a_{10} + a_{11}y + 2a_{20}x + 2a_{21}xy + a_{12}y^2 + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial f / \partial x)_y|_{0,0} = a_{10}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$\begin{aligned} f(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + \dots \end{aligned}$$

$$(\partial f / \partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial f / \partial y)_x|_{0,0} = a_{01}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$\begin{aligned} f(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + \dots \end{aligned}$$

$$(\partial f / \partial y)_x = a_{01} + a_{11}x + 2a_{02}y + a_{21}x^2 + 2a_{12}xy + \dots$$

$$(\partial^2 f / \partial x \partial y) = a_{11} + 2a_{21}x + 2a_{12}y + \dots$$

$$x = 0, y = 0 \text{ gives } (\partial^2 f / \partial x \partial y)|_{0,0} = a_{11}$$

Power Series in Several Variables

For a two-variable function $f(x, y)$, we can write

$$\begin{aligned} f(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + \dots \end{aligned}$$

In general,

$$a_{nm} = \frac{1}{n! m!} \left(\frac{\partial^{n+m} f}{\partial^m y \partial^n x} \right) \bigg|_{0,0}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x, y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y .

Consider $f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 + \dots$

$$f(0,0) = e^0 = 1 = a_{00}$$

$$(\partial f / \partial x)_y \Big|_{0,0} = e^{x+y} \Big|_{0,0} = e^0 = 1 = a_{10}$$

$$(\partial f / \partial y)_x \Big|_{0,0} = e^{x+y} \Big|_{0,0} = e^0 = 1 = a_{01}$$

All the derivatives are equal to e^{x+y} and

$$a_{nm} = \frac{1}{n! m!} \left(\frac{\partial^{n+m} f}{\partial^m y \partial^n x} \right) \Big|_{0,0} = \frac{1}{n! m!}$$

Example 10.13

Find the first few terms of the Maclaurin series representing the function $f(x, y) = e^{x+y}$. Compare your result with the product of the Maclaurin series representing e^x and the series representing e^y .

$$e^{x+y} = 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \dots$$

The Maclaurin series representing e^x is

$$e^x = 1 + x + (1/2!)x^2 + (1/3!)x^3 + \dots$$

Similarly,

$$e^y = 1 + y + (1/2!)y^2 + (1/3!)y^3 + \dots$$

Hence, the product of e^x and e^y is

$$\begin{aligned} e^{x+y} &= (1 + x + (1/2!)x^2 + \dots)(1 + y + (1/2!)y^2 + \dots) \\ &= 1 + x + y + xy + (1/2!)x^2 + (1/2!)y^2 + \dots \end{aligned}$$